

The Production and Consumption of Commodities in the Amazon Rubber Industry

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May 8th 2026

Introduction

Until 1913, the Amazon was by far the largest producer of rubber for the world market. During the Amazon Rubber Boom the massive pressures of producing rubber for the emerging automotive industry transformed life in the Amazon. Investments in infrastructure connected the relatively isolated region to global markets, while tens of thousands of people migrated to the region looking for work. At the same time, the (under)development of Amazonian industry followed a very different trajectory than the rise of the automobile industry in the US. Despite the wealth generated by the exploitation of rubber, there was little long-term, stable investment into production, and the Amazon economy collapsed in the 1920s with the rise of cheaper rubber from plantations in Asia. Additionally, the workers in the Amazon experienced high mortality rates and treacherous working conditions, with little of the stability afforded workers in Ford's automotive factories.

The forms of labor exploitation in the Amazon differed significantly from those which prevailed in industrialized countries. Wage-labor was relatively uncommon in the Amazon, and rubber traders instead coerced tappers to work through usurious debt contracts, and sometimes outright force. Meanwhile, wealth disparities were enormous - as some tappers lived close to starvation, the urban centers of the Amazon boasted spectacular wealth and imported customs and commodities from Europe. However, despite the difference in living and working conditions

between the rubber workers of the Amazon and those in automotive factories in the US and Europe, they were two poles of a single system. The extreme exploitation of the rubber workers, and correspondingly the cheapness with which rubber was procured, was a crucial condition of profitability for the automotive industry, and thereby subsidized the higher wages granted to Ford's factory workers. At the same time, it was the success of the automotive industry that drove the growth of the rubber industry and demanded commodity production from a region whose characteristics made wage-labor near impossible. Each region, and the actors within them, formed part of a totality - and each is impossible to analyze without the other.

This chapter will analyze social reproduction in the rubber industry of the Amazon as a constitutive element of the capitalist world-system. Social reproduction is a complex dynamic with several interlocking parts. Following the recent body of scholarship known as Social Reproduction Theory,¹ we will be looking at the way in which individuals are produced and reproduced as laborers within a commodity producing society - what Marx calls the "restoration, renewal and refreshment of the vital forces" of the worker.² In other words we are concerned with the conditions that bring laborers to the factory door every morning - or in this case to the rubber tree. This consideration is important for capital in both the short- and long-term. Reproducing a labor force means ensuring not only that workers are able to rest and renew themselves as working subjects from day-to-day, but also ensuring that a new stream of workers is ready and able to replace existing workers who retire, leave, die, or are incapacitated through injury or disease. This reproduction of labor is most often effected generationally, through the

¹ Tithi Bhattacharya, ed., *Social Reproduction Theory: Remapping Class, Recentering Oppression* (Pluto Press, 2017).

² Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976). 376.

rearing of children in workers' homes, but also through institutions of migration, labor conscription and enslavement.³

Labor is not the only element of capitalist society reproduced socially, and so following Kirstin Munro's critique of Social Reproduction Theory we will also be looking at the way in which the consumption practices (or 'household production') of workers in the rubber industry contributed to the valorization of capital and thereby to the renewal of the conditions under which capital accumulates.⁴ The way in which commodities are consumed by workers and private persons both structures and is structured by the interplay of global and local markets. In many ways the household is both the starting point and end point of the cycle of capital accumulation - it is the well from which labor is drawn, and the place in which the exchange-value of commodities is finally realized as their use-values are consumed and demand is renewed.

The lives of the inhabitants of the Amazon - both established communities and newly arrived migrants and refugees - were transformed by the rubber industry. Settlers looking to exploit the natural resources of the region destroyed Indigenous communities and their ways of life. This was accelerated by the push to extract rubber for the global market. Not only did the capitalists and traders press Indigenous communities into serving as labor for the global automotive commodity chain, but they also interrupted indigenous patterns of reproduction - especially practices of food cultivation - and replaced them by commodity consumption. In this way the indigenous communities were incorporated into the world economy as both a labor

³ Lise Vogel, *Marxism and the Oppression of Women: Toward a Unitary Theory*, Revised Edition, Historical Materialism Book Series 45 (BRILL, 2013). 157-163.

⁴ Kirstin Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," *Science & Society* 83, no. 4 (2019): 451-68.

source and as a market for finished commodities from the Global North, such as canned food, tools, medicine, and cloth.

This dynamic resembles the process of proletarianization as discussed in Marxist literature. Indigenous people's ability to reproduce themselves directly from the land was interrupted as they were dispossessed of their commons, and in its stead a commodity market was established. Only through the exchange of their labor-power could they access the social product necessary for their survival. In this sense, they were 'freed' from their connection to the land. In many cases they entered into contracts with their employers and were also 'free' in the eyes of the law. But for the most part this freedom was a legal fiction, and they were held in debt-bondage and servitude not only by the impersonal compulsion of the market but by force. At any rate, they were not free waged workers, and their conditions have been described by historians of the region as 'pre-capitalist' formations, residual and anachronistic social structures held over from the colonial period.⁵

This chapter re-evaluates the characterization of the Amazonian rubber industry as pre-capitalist. I agree that the rubber tappers were unfree, and that their conditions differed greatly those of wage-laborers in the core. Indeed their laboring circumstances could not have resembled Fordist factories less. But I disagree strongly that their conditions were in any way out of place in the capitalist world. Their conditions were the same as those experienced by the vast majority of the Global South after being incorporated into the world economy. They were what world systems analysts call part lifetime proletarians: "laborers who depend on non-market sources for

⁵ Barbara Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920," in *Proletarians and Protest: Studies in Class Formation*, ed. Michael P. Hanagan and Charles Stephenson (Greenwood Press, 1986).

part of their livelihood rather than solely on wages."⁶ They combined income from wages, illicit sales of rubber hidden from their bosses, restricted subsistence labor, and off-season work in the urban areas of the Amazon in order to survive. Their living and laboring conditions benefited global capital in two ways - to produce cheap raw materials for growing industry, and to serve as a market for finished commodities. In both their labor inputs and in the patterns of their private lives, the conditions of rubber workers in the Amazon were not just dictated by the capitalist world-system, but were constitutive of it.

The Amazon Rubber Boom

Rubber has been used by humans for centuries. When Europeans arrived in the Americas in the fifteenth century, they witnessed Indigenous communities use rubber to make balls, bottles, syringes, and to waterproof cloth and shoes.⁷ However, despite its elastic and water-repellent properties, it was of little use to Western industrialists as it became sticky in the heat and rigid in the cold - its industrial application was largely limited to raincoats. In the 1830s Charles Goodyear developed and then perfected the process of vulcanization which stabilized rubber and made it more suitable for industrial and consumer applications.⁸ Vulcanized rubber proved quite useful in various industries, and would receive a massive boost in 1888 with John Dunlop's pneumatic bicycle tire and the creation of a rubber tire industry.⁹ Within 5 years, "the only tires [in use] on self-propelled vehicles [were] either solid rubber or pneumatic rubber tires;

⁶ Ravi Arvind Palat, "Dependency Theory and World-Systems Analysis," in *A Companion to Global Historical Thought* (2014). 377.

⁷ Austin Coates, *The Commerce in Rubber: The First 250 Years* (Oxford University Press, 1988).

⁸ Herbert Morawetz, "History of Rubber Research," *Rubber Chemistry and Technology* 73, no. 3 (2000): 405–26.

⁹ Burgess Darrow, "Pneumatic Tires - Old and New," *SAE Transactions* 27 (1932): 438–44.

no other type seems as the present time likely to engage the serious attention of engineers."¹⁰ It caught the attention of capitalists as well as engineers, and the number of firms producing pneumatic tires in the US grew from 1 in 1896 to 157 in 1917 and reaching a peak of 278 in 1922.¹¹ Driven by this growth in tire manufacturing, US rubber imports rose from 20,308 long tonnes in 1900 to 157,371 in 1917 and 301,499 in 1922. Imports would continue to grow, reaching half a million in the 1930s.

Prior to the development of rubber tires, and for the first few years of the new industry's growth, the majority of the world's rubber was sourced from the Amazon. However, by the third decade of the twentieth century the Amazonian rubber industry had collapsed and was replaced on the market by Asian rubber, especially from Malaya and Sumatra. In fact the automotive tire industry, though it initially boosted demand for Amazonian rubber, eventually spelled the end of Amazonian dominance in the industry. The exponential growth in demand for natural rubber could not be met by the inefficient harvesting methods used to tap the wild rubber trees that dominated the Amazon. Meanwhile both the rubber tappers and the ecology of the Amazon resisted the attempts at rationalization that would be necessary to produce rubber on a large scale. The sluggishness of Amazon rubber production gave way to a more efficient plantation system in Asia that lacked such labor and ecological issues.¹²

Various dates are given for the Amazon Rubber Boom, but we can generally consider it to have lasted from ~1870 to 1913.¹³ The first date is hard to establish with precision as rubber

¹⁰ P. W. Litchfield, "PNEUMATIC TIRES," *Transactions (Society of Automobile Engineers)* 8 (1913): 208–23.

¹¹ Guido Buenstorf and Steven Klepper, "Submarket Dynamics and Innovation: The Case of the US Tire Industry," *Industrial and Corporate Change* 19, no. 5 (2010): 1563–87.

¹² The rise of plantation labor in Asia will be the subject of Chapter 5.

¹³ Bradford L. Barham and Oliver T. Coomes, *Prosperity's Promise: The Amazon Rubber Boom and Distorted Economic Development*, *Dellplain Latin American Studies* 34 (Westview Press, 1996); John

production statistics are scarce in the 19th century, but the end date is more decisive. Although the economy of the Amazon was dominated by rubber production into the 1920s, 1913 was the last year in which the quantity of wild rubber harvested in the Amazon was greater than that grown on rubber plantations in Malaya.¹⁴ During this boom period, practically all of the rubber produced for industrial purposes was harvested from wild rubber-trees by roaming tappers. Because of the peculiarity of the method required to harvest wild rubber as well as the remoteness of the rubber trees from industrial infrastructure, both the means of labor control and the patterns of capital investment differed from those seen in the industrialized US.

Two of the elements that differentiate the Amazonian rubber economy vis-a-vis the industrialized economies of the North are a lack of a stable and waged labor force, and a lack of capital investment aimed at increasing the productivity of labor. Each of these contributed to the absence of stable long-term industry in the Amazon - its underdevelopment. In theorizing the unique characteristics of industry in the Amazon, Stephen Bunker coined the phrase 'mode of extraction' to indicate an economic logic different from the 'mode of production' which prevailed in the economies of the US and Europe. For Bunker, "[t]he crucial difference between production and extraction is that the dynamics of scale in extractive economies function inversely to the dynamics of scale in the productive economies to which world trade connects them."¹⁵ In other words, whereas in productive economies greater output is brought about by an increase in the productivity of labor through its combination with an investment in technology, in extractive

Tully, *The Devils Milk* (NYU Press, 2011), JSTOR, <http://www.jstor.org/stable/j.ctt9qfjqp>; Barbara Weinstein, *The Amazon Rubber Boom, 1850-1920* (Stanford University Press, 1983).

¹⁴ P. W. Barker and E. G. Holt, *Rubber Statistics 1900-1937: Production, Absorption, Stocks and Prices*, no. 181, Trade Promotion Series (U. S. Department of Commerce, 1938).

¹⁵ Stephen G. Bunker, *Underdeveloping the Amazon: Extraction, Unequal Exchange, and the Failure of the Modern State* (University of Illinois Press, 1985). 25

economies a greater yield is effected by an increase in the amount of labor employed to exploit each unit of a dwindling natural resource. A greater extractive output corresponds to a fall in the productivity of labor. Because of this, "when extractive systems respond to increased external demand, they tend to impoverish themselves (1) by depleting non-self-renewing resources or (2) by exploiting self-renewing resources beyond their capacities for regeneration."¹⁶ The extreme exploitation of resources, and especially the high labor inputs, raise the cost of extraction so high as to stimulate cultivation or synthetic production of the natural resource elsewhere.

This model perhaps is a better fit for the Amazon's other industries than for rubber. There are several types of plants that produce quality rubber in commercial quantities. The two most prominent in the Americas are *Castilla Elastica* which grows as far north as Mexico, and *Hevea Brasiliensis*, which grows only in the Amazon. Of these, *Castilla* was the first to be harvested. It provided the rubber used by Mesoamerican peoples and was the dominant source of rubber through the first decades of the Amazon Rubber Boom. In order to extract natural latex from *Castilla*, harvesters would usually cut down the entire tree and manually extract the sap from the bark as the trunk decays. By contrast, *Hevea Brasiliensis* has a capillary system that bears rubber near the bark. This system can be exposed through shallow cuts, and the tree can be repeatedly and productively tapped.¹⁷ Indeed, Bunker notes that *Hevea* "will yield indefinitely if properly scarred and tapped."¹⁸

Both sources of rubber are renewable, but it is only the first which fits into Bunker's schema for impoverishment by over-exploitation. *Castilla* was more prominent in the upper

¹⁶ Bunker, *Underdeveloping the Amazon*. 25

¹⁷ O. F. Cook, "Rubber Production from *Castilla* and *Hevea*," *Science* 85, no. 2208 (1937): 406–7, <https://doi.org/10.1126/science.85.2208.406>.

¹⁸ Bunker, *Underdeveloping the Amazon*. 66

reaches of the Amazon, and sources indicate that by the start of the twentieth century it was sparse from over-harvesting.¹⁹ Production from *Hevea*, which accounted for the vast majority of Amazonian rubber, was limited not by over-harvesting but by *under-harvesting* owing to a dearth of labor. In one of the earliest works analyzing the Amazonian rubber industry Charles Akers claims that the natural stock of rubber trees "for practical purposes may be considered inexhaustible; but the supply of latex for the manufacture of rubber for export is controlled by the lack of any large laboring population."²⁰ In the 90s, Coomes and Barham reaffirm this analysis, stating that "the scarce factors of production in the wild rubber industry were labor and capital, in contrast to rubber trees which were relatively abundant."²¹

That the development of a region so rich in natural resources could be stymied by a scarcity of labor and capital in an era of unprecedented advances in infrastructural advancement and labor mobility is baffling. Indeed it would be incorrect to assert that capital and labor did not flow into the Amazon. There were concerted efforts by Brazil to import labor, both from abroad and from other regions inside the country, into the Amazon, many of them quite successful.²² Furthermore, the urban centers of the Amazon saw massive expansion with Manaus boasting an "excellent system of waterworks, an efficient garbage collection and disposal system, electricity, telephone service, handsome public buildings, and comfortable private residences" - all

¹⁹ Cook, "Rubber Production from Castilla and Hevea."

²⁰ Charles Edmond Akers, *The Rubber Industry in Brazil and the Orient* (Methuen & Co. Ltd., 1914). 8

²¹ Oliver T. Coomes and Bradford L. Barham, "The Amazon Rubber Boom: Labor Control, Resistance, and Failed Plantation Development Revisited," *Hispanic American Historical Review* (Durham) 74, no. 2 (1994): 231–57.

²² For an example of failed foreign settlements in the Amazon, see Lawrence F. Hill, "Confederate Exiles to Brazil," *The Hispanic American Historical Review* 7, no. 2 (1927): 192–210. The more successful system of internal migration will be dealt with below.

indications of significant capital investment in the region.²³ But the influx of people and cash did not alleviate the problems of labor and *productive* capital investment. This in part because of the ways in which both money and labor were employed in the Amazon. The labor situation will be dealt with in following sections, but it is instructive to treat here the growth of the rubber boom towns.

During the rubber period, Manaus and Belem grew enormously. Between 1872 and 1920, Manaus grew from 29,000 inhabitants to 75,000, whereas the population of Belem increased from 61,000 to over 230,000 in the same period.²⁴ This rapid expansion in population was accompanied by the aforementioned infrastructural projects, but also more by conspicuous displays of wealth, such as the notorious Amazonas Theater in Manaus. This theater, which was completed in 1896, was made almost entirely from imported materials:

For its construction not only architects, builders, painters, decorator and sculptors were brought from Europe, but also various materials such as Carrara marbles, Murano chandeliers, pieces of worked iron from England, French tiles and European wood for the floor. In the internal structure, the steel dome was manufactured in Belgium.... The dome roof is composed of 36 thousand pieces of enameled ceramic pottery and glazed tiles, coming from Alsace.²⁵

In addition, there was a robust market for commodities from Europe and the United States, as advertisements listed for sale such items as "Smith and Wesson revolvers, Collins machetes, Victor gramophones, Walk-over shoes, Parfum Lubin, Omega watches, Black and White whiskey, Booth gin, Scandinavian butter, Fussell condensed milk, Otto motors, Liqueurs Bardinet, Sun typewriters, Deutz automobiles, Underwood typewriters, White Label whiskey,

²³ E. Bradford Burns, "Manaus, 1910: Portrait of a Boom Town," *Journal of Inter-American Studies* 7, no. 3 (1965). 401.

²⁴ IBGE, *Censo Demográfico*, 1872; and IBGE, *Censo Demográfico*, 1920.

²⁵ Sampaio Maria S.. M., "Amazonas Theater Architectural Construction and Restorations History," paper presented at 12th International Conference on Structural Analysis of Historical Constructions, 2020, https://www.scipedia.com/public/M. Sampaio_2021a. 2.

etc." There were also French restaurants, sports clubs, and other trappings of European life, but it is specifically the importation of commodities from the core industrial countries that is significant for this study.

Manaus is held up as a prototypical boom town, where quick fortunes were made in the day by the exploitation of land and people and then lost at night in the bordellos and bars. The conspicuous consumption on display in Manaus is sometimes considered a weakness, which ultimately undermined the possibility for stable development - why was this money not spent on infrastructure, or on rationalizing production?²⁶ This local focus misses the crucial role that such consumption played on a global scale. Commodity imports are not unidirectional: while commodities flowed into Brazil, the surplus from the rubber industry flowed outward to the United States and Europe. Capitalism is not merely a system of commodity production, but of commodity consumption, and by the creation of a market for finished goods in the boom towns of the periphery, the value produced in the core can be realized.

In short, the Amazon region benefited the industrial Global North in two ways - both in the production of cheap raw materials, and in the consumption of finished goods. Both roles undermined the possibility of long-term stable production, as the labor turnover rate (as well as the death rate) in harvesting was high, and the surplus earned was more often spent on conspicuous displays of wealth than invested into increasing the productivity of labor. The following sections will examine the labor relations and capital investment dynamics in Amazonian Rubber during and following the boom, further highlighting the Amazon's failure to transition from what Bunker calls an extractive economy to a productive one.

²⁶ See the section *Weaknesses in the Brazilian Industry* in Randolph R. Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945," *The Business History Review* 51, no. 3 (1977): 341-66.

The 'Free' Laborer

The procurement of labor for commodity rubber production in the Amazon differed in key respects from the process of proletarianization seen in industrial companies, especially in its failure to establish a body of free wage-laborers. But under capitalism, free labor takes on a second, ironic meaning. The proletarian is "free in the double sense that as a free individual he can dispose of his labour-power as his own commodity, and that, on the other hand, he has no other commodity for sale, i.e. he is rid of them, he is free of all the objects needed for the realization of his labour-power."²⁷ In other words, workers under capitalism are 'freed' from the ability to reproduce themselves by the dissolution of their traditional social ties and community relationships. This 'freedom' comes about as a result of a historical process of dispossession whereby workers are cut loose from the land (or from other social structures which sustain them) and forced into a market for labor. Though not formally wage-workers, the rubber tappers in the Amazon were free in this second, ironic sense.

In the nineteenth century, several major events undermined the social and economic relations that structured Brazilian society, and left hundreds of thousands of workers without their established means of survival. Chief among these were the abolition of slavery and the *Grand Seca* that devastated the northeast. These two events each set loose some portion of the population which had previously been bound in a customary social hierarchy. Unlike their fellow workers of the industrialized Global North, however, neither the climate refugees from *Grand Seca* nor the formerly enslaved Africans became wage workers. They were rather recruited to work under conditions of debt peonage as *seringueiros*, or rubber workers, in the Amazon where

²⁷ Marx, *Capital*, vol. 1. 272-273.

market penetration was shallow and the flow of those commodities necessary for survival was tightly controlled by a group of middlemen-traders, or *aviadores*.

In the rubber fields, they joined the existing population of peoples Indigenous to the Amazon whose traditional ways of life had been similarly disrupted. Through hundreds of years of colonization, during which colonists massacred them, displaced them, and pressed them into involuntary servitude, Indigenous communities had been cut off from the land. By being compelled to labor not for their own subsistence but for a series of colonial masters, from Jesuit missionaries to state agents and settlers, these communities were forced to become dependent on a flow of commodities from outside the Amazon as their metabolism with the forest was broken. This section will trace how each of these three groups - Indigenous peoples, climate refugees from the *sertão*, and formerly enslaved Africans - were 'freed' from the social structures that maintained them and were recruited into the rubber trade.

The Indigenous Peoples of the Amazon

When rubber found a market on a global scale, the Indigenous peoples of the Amazon were among the first to be pressed into the production of rubber as a commodity. This was not a new dynamic, as colonial powers (and later the Latin American states) had exploited the inhabitants of the Amazon since the arrival of Europeans to South America at the turn of the 16th century. Indeed "exploited" is too gentle of a word, as the European powers and their Latin American successors committed genocide against the Indigenous people. Massacres, enslavement, forced displacement, and ecocide depopulated the Amazon by 80%, from around 5 million in 1500 to around a million four hundred years later, falling further to around 200,000 in

1980.²⁸ Rubber barons themselves participated in the genocide of Indigenous Amazonians, such as in the Putumayo region of the upper Amazon where state presence was more limited. But in much of the lower Amazon the demographic collapse of Indigenous peoples undermined the nascent rubber industry.

The combination of declining population and an ecology unfit for labor-intensive plantation development led early European settlers to organize "their economic penetration of the region during the seventeenth and eighteenth centuries around highly mobile groups of Indian gatherers who dispersed throughout an area to extract natural substances such as turtle oil, spices, hardwoods, vegetable oils, and cacao beans."²⁹ The collectors were sometimes paid a wage, but more often given tools, cloth, or trinkets in exchange for the goods they gathered. This differed dramatically from the economic foundation of the rest of Brazil which developed plantations on the basis of the labor of enslaved Africans. However, the system of voluntary exchange with the Indigenous population was not sufficient to meet the demands of the Europeans, and so the colonists began "a series of slaving expeditions in the major areas of Indian settlement in order to assure themselves of an adequate labor force."³⁰ Indigenous communities resisted these raids, but Indigenous resistance was often used as a pretext for more violent retaliation. The force of the colonists was overwhelming. The slaving raids accelerated the collapse of Indigenous communities and by the time that the rubber industry began to grow in the nineteenth century, the labor requirements were in considerable excess of the population.

During the 17th and 18th centuries, there was another group influential in the capture and exploitation of Indigenous labor in the Amazon: the Jesuits. Often portrayed as beneficent

²⁸ Chris C. Park, *Tropical Rainforests* (Routledge, 1992). 108.

²⁹ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 10

³⁰ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 11

towards the population of the Amazon, the Jesuits were in fact instrumental in justifying a system of forced labor, relying on the colonial principles of both Just War and *resgate* (rescue) to determine who could be legally enslaved.³¹ The Portuguese crown invested in founding mission villages in the region, which would be responsible for both the spiritual well-being of the nearby communities and for establishing a system of control over Indigenous labor. The majority of the surrounding Indigenous groups would be pressed into service at the mission itself such as tending to foodstuffs and "conduct[ing] other Indians down from their jungle homes to designated mission villages."³² The Indigenous communities, as well as their relations of reproduction, were destroyed, and the people were forced to labor for the reproduction of the missions themselves - producing food and procuring more labor. Those people not pressed into service at the missions were made available as laborers first the government and finally to private settlers.

The Jesuits and settlers clashed over control of the Indigenous population of the Amazon, with the missions periodically disallowing colonists access to their labor, allegedly due to the colonists' mistreatment of the workers. In 1759 the Portuguese sided with the colonists and expelled the Jesuits from Brazil (and other areas of the Portuguese Empire), but "the collapse of the Amazonian mission system did little to resolve the settlers' economic troubles, since the fundamental problem was the prolonged and precipitous decline of the Indigenous population."³³ Access to labor was to be a major issue for extractivist ventures in the Amazon, and with a

³¹ A. J. R. Russell-Wood, "Introduction: New Directions in Bandeirismo Studies In Colonial Brazil," *The Americas* 61, no. 3 (2005): 353–71.

³² Colin M. MacLachlin, "The Indian Labor Structure in the Portuguese Amazon, 1700-1800," in *Colonial Roots of Modern Brazil: Papers of the Newberry Library Conference*, ed. Dauril Alden (University of California Press, 1973). 201

³³ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 12

dwindling local population, it would have to come from somewhere else. The major sources would be enslaved or formerly-enslaved Africans, and climate refugees from the drought-stricken region of Northeast Brazil.

The Abolition of Slavery in Brazil

Shortly before the Jesuits were removed from Brazil, the Portuguese Crown shifted its sights from Indigenous to enslaved African labor in the Amazon. In 1755, "all forms of Indigenous slavery were abolished" and the Grão-Pará and Maranhão General Trading Company was established "to promote the importation of enslaved Africans to the Amazon."³⁴ By the end of the 18th century, the Company had imported almost 20,000 enslaved people into the Amazon region from other parts of Brazil. Although this number was too small to fill all of the labor needs of colonists in the region, it had a significant impact on the labor relations in the Amazon and the demographic composition. During the 18th and 19th century, the distinction between such racial categories as "Indian" and "Black" became permeable, as did the distinction between free and unfree labor. Today, a majority of the residents of the Amazon (at least 65% in each Amazonian state) identify as 'Black'.³⁵

During the Atlantic slave trade Brazil was the largest importer of enslaved Africans, serving as the final destination for over 5 million persons.³⁶ They were employed first in sugar

³⁴ Adalberto Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon," trans. Bryan Pitts, *International Review of Social History* 62 (2017). 26.

³⁵ David Pereira Junior, "AMAZON REGION AND BLACKNESS: NOTES TO UNDERSTAND BLACK INVISIBILITY," *Journal of the Research and Training Center*, 2025, 117–39.

³⁶ For a survey of the early critical historiography of Brazilian Slavery, see Richard Graham, "Brazilian Slavery Re-Examined: A Review Article," *Journal of Social History* 3, no. 4 (1970): 431–53. For a survey of more recent literature, see Rafael Marquese and Ricardo Salles, "Slavery in Nineteenth-Century Brazil: History and Historiography," in Dale Tomich, ed., *Slavery and Historical Capitalism during the Nineteenth Century* (Lexington Books, 2017).

cane plantations, and later in mining and coffee operations. Mortality in these fields was very high - on the sugar plantations, the death rate of the enslaved exceeded their birth rate.³⁷ In theory it was possible for enslaved people to purchase their freedom, and at the death of an enslaver there was a "generalized practice of manumissions, which originated an enormous population of ex-slaves."³⁸ However, the plight of those released from slavery was so unstable that "[n]ot only did the freedman have less security than the slave and generally become the dependent of his exmaster, but he was often used in the most dangerous work where the masters feared to trust their slaves."³⁹

For individuals of either African or Indigenous ancestry in the Amazon - whether enslaved or not - labor was subject to various types of coercion. As they could still be compelled to work for the state or private citizens, the freedom that had been granted to the Indigenous population in 1755 was, "not the freedom to come and go as one pleased, but was rather more akin to a limited concession that could be curtailed and conditioned at any time."⁴⁰ In 1798 a law was passed modifying the post-slavery system of labor for Indigenous people that "added a fundamental innovation to previous Indigenous policy by imposing compulsory service on the entire free, non-white population, including blacks, *mamelucos*, and *cafuzos*."⁴¹ The liberty of free Africans in the Amazon thus became likewise subject to the labor needs of both the state and colonists.

³⁷ Kit Sims Taylor, "The Economics of Sugar and Slavery in Northeastern Brazil," *Agricultural History* 44, no. 3 (1970): 267-80, JSTOR.

³⁸ Marquese and Salles, "Slavery in Nineteenth-Century Brazil: History and Historiography." 145.

³⁹ Graham, "Brazilian Slavery Re-Examined: A Review Article." 452-453

⁴⁰ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 27.

⁴¹ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 29.

Naturally, both the Indigenous peoples and the Africans resisted the oppressive system of racialized labor imposed on them. Between 1835 and 1841 a civil war was waged across northern Brazil called the *Cabanagem*, the longest and largest rebellion in the history of Imperial Brazil.⁴² The elites in Brazil perceived this war as a general uprising of the *gente de cor*, or people of color. The *Cabanagem* left as many as 30,000 dead and resulted in "the widespread disorganization of the rural labor force" as "enslaved Africans and quasi-enslaved Indians joined the guerrilla bands which roamed the region."⁴³

In the aftermath of the *Cabanagem* the Brazilian state doubled down on attempts to pacify racialized labor in the Amazon valley. In the 1838 Pará created a labor corps, the *Corpo de Trabalhadores*, to press the population of color into service for a nominal wage. These corps engaged in the forced resettlement of the Indigenous population to towns where labor needs were higher, and along with the legal framework that established them "represented a virtually unlimited expansion of the right of any civilian, military, or state-sanctioned private authority to exploit the involuntary labor of Indians, blacks, and *mestiços*."⁴⁴ It is interesting to note that these labor corps considered rubber collectors to be vagrants, as they "lived in complete independence," and would conscript them to work in more settled industries.⁴⁵

However, this particular system of involuntary servitude was short lived, and the workers corps were eventually abandoned in the state of Pará in the late 1850s. During this time, Brazil was entered its abolition period, but more importantly rubber began at this period to dominate the

⁴² David Cleary, "'Lost Altogether to the Civilised World': Race and the Cabanagem in Northern Brazil, 1750 to 1850," *Comparative Studies in Society and History* 40, no. 1 (1998): 109–35.

⁴³ Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920." 59.

⁴⁴ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 31.

⁴⁵ Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920." 59.

Amazon and its need for 'independent' workers trumped the concerns of the less lucrative Amazonian agrarian industries. However, the abolition of the workers corps did not mean the freedom of labor, especially for racialized workers and the formerly enslaved.

Although it persisted in modified forms of coerced labor, legal slavery was formally abolished in all of Brazil in 1888. As a legal institution it had been crumbling for decades. As mentioned earlier, the enslavement of Indigenous people was outlawed in 1755, and in 1871 Brazil passed the Rio Branco Law, or the Law of the Free Womb, which granted free status to all children born to enslaved women. In March 1884, four years before the nationwide decree, slavery was ended in the state of Ceará, whose economy was more dependent on the labor of peasants than of enslaved Africans.⁴⁶ Four months later, Amazonas, which had about 2% of the enslaved population of its neighboring Amazonian state Pará, followed suit.⁴⁷

The abolition of slavery in Brazil was slow and incomplete. Brazil was the last country in the Americas to grant freedom to the victims of the Atlantic slave trade, and they did so without compensation. Rather than providing the newly freed people with the means to participate fully in civic and economic life, the liberation of Africans in Brazil "resulted in either (1) the marginalization of former slaves from capitalist production or (2) the incorporation of former slaves into new, race-based regimes of coerced labor."⁴⁸

⁴⁶ Erick Matias, "March 25, 1884: Ceará Abolishes Slavery Before the Golden Law and Enters Brazilian History," *O Maringá*, March 25, 2026, <https://omaringa.com.br/en/coluna/datas-historicas/25-de-marco-de-1884-ceara-abole-a-escravidao-antes-da-lei-aurea-e-entra-para-a-historia-do-brasil/>.

⁴⁷ Gabriel Abreu and Bruno Pacheco, "137 Years Ago, AM Signed the Slave Liberation Charter: 'We Are Still the Protagonists of Our Own Freedom,'" trans. Bruno Sena, *Revista Cenarium*, July 12, 2021, <https://agenciacenarium.com.br/137-years-ago-am-signed-the-slave-liberation-charter-we-are-still-the-protagonists-of-our-own-freedom/?lang=en>.

⁴⁸ David Baranov, *The Abolition of Slavery in Brazil: The "Liberation" of Africans Through the Emancipation of Capital*, Contributions in Latin American Studies 17 (Greenwood Press, 2000). 7.

Common to all of the racialized labor regimes in Brazil - enslaved, coerced, or formally free but precarious - was the expectation that people of color in the Amazon were obligated to provide labor to meet the needs of capital and the state. Just as importantly, through the forced resettlement of Indigenous and enslaved people as well as the economic legacy of slavery, the ability of Brazil's racialized underclass to sustain itself by its own labor had been thoroughly undermined. For both Indigenous and African communities, "[t]he abolition of slavery in Brazil was, in fact, premised on the creation of social conditions that assured the former slave population's uninterrupted servitude, guaranteed by the heavy hand of extramarket structures and institutions."⁴⁹ During the rubber trade, both of these populations were incorporated into the growing class of rubber tappers and subject to new forms of both economic and extra-economic compulsion. The growth of the rubber industry was so great, however, that the Black and Indigenous residents of the Amazon could not meet the demand for labor. The majority of rubber workers would have to be pulled from elsewhere in Brazil - the peasant population of the northeast.

The Grand Seca

The northeast of Brazil is dominated by the *sertão*, a semi-arid and sparsely populated region visited periodically by great droughts. The turn of the century saw a series of especially bad droughts, the worst of which was the *Grand Seca* of 1877. This drought and those which followed saw a fundamental reconfiguration of the *sertão*'s social fabric. Between 1877 and 1919, two million people in the northeast died from the droughts and their effects. In addition to this, tens of thousands of the inhabitants of the *sertão* left their homes to look for work and

⁴⁹ Baranov, *The Abolition of Slavery in Brazil*. 168.

subsistence in other parts of Brazil, principally working as rubber harvesters in the Amazon. The already precarious conditions of life and labor in the northeast were so decisively undermined that "the infamous drought of 1877 ... definitively ended slavery in Ceará, an event which anticipated the emancipation of all slaves throughout Brazil in 1888."⁵⁰

Mike Davis argues that drought "always has a manmade dimension and is never simply a natural disaster."⁵¹ Indeed, although the droughts in the *sertão* are linked to the semi-regular occurrence of the El Niño in the Pacific, their human cost is a result not merely of climatic caprice but is inflamed by various human factors, from the development of infrastructural projects (or lack thereof) to the prevailing property relations in affected regions. In the nineteenth century, Brazil was a major producer of both coffee and sugar, but these were mainly limited to the south of the country. In Ceará, the state most affected by the *Grand Seca*, the population relied on cattle and cotton produced for export. Although better suited to the arid climate of Ceará than other industries, cattle raising cannot be said to have thrived there. Davis notes that cattle were "forced out of the *zona de mata* by the sugar boom," and settled in the *sertão* as a sort of last resort. Whereas in much of the world a hectare of ranch land can support 1-2 cattle, in northeast Brazil, a "typical ranch of 1,000 hectares, for example, might sustain only 50 scrawny cattle; and even the biggest *fazendas* (10,000 hectares or more) rarely pastured herds larger than 1,000."⁵²

⁵⁰ Reuben H. Brooks, "HUMAN RESPONSE TO RECURRENT DROUGHT IN NORTHEASTERN BRAZIL" *The Professional Geographer* 23, no. 1 (1971): 40–44, <https://doi.org/10.1111/j.0033-0124.1971.00040.x>. 42

⁵¹ Mike Davis, *Late Victorian Holocausts: El Niño Famines and the Making of the Third World* (2001; Verso, 2017). 20

⁵² Davis, *Late Victorian Holocausts*. 405

Cotton began to eclipse cattle in the eighteenth century as droughts weakened cattle herds, especially during the American Civil War which disrupted the global cotton market. Large plantations combined cotton and cattle, but they crucially also relied on subsistence agriculture to maintain the working population. In the absence of a robust commodity market for the means of subsistence, the large *fazendas* had to be able to maintain not just the workers, but their families as well. Well into the 20th century, the population of the *sertão* "employs the same food technology as its Indian predecessors, subsisting primarily on manioc and maize and secondarily on sweet potatoes, beans, kidney beans, pumpkins, peanuts, pineapple, pepper, gourds, and tobacco."⁵³

The right to cultivate subsistence crops was a custom granted to the *moradores* working the *fazendas* as a part of a system of mutual obligation between the owners and the workers who lived on and worked their land. While the owners could dictate land use, demand labor for upkeep around the *fazenda*, fire unruly workers, and had a right to any cotton produced by the *moradores*, they had several obligations to the *moradores* as well. *Moradores* and their families were entitled to housing, hunting rights, and could fish in the dammed ponds, although they did not have "an automatic right to the use of this water for irrigation purposes."⁵⁴ The workers also did not have the right to ask a bank for a loan, and had to go through the owners.

Crucially, however, unlike peasants in feudal societies, the *moradores* and their families were not legally bound to the land. Mobility was a constant part of life in the *sertão*. Bachelors were the most mobile and could attach themselves to *moradore* families, but once a nuclear family was established they were generally expected to "settle into a tenant relation with an

⁵³ Bernard J. Siegel, "Migration Dynamics in the Interior of Ceará, Brazil," *Southwestern Journal of Anthropology* 27, no. 3 (1971). 235

⁵⁴ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 251

owner."⁵⁵ Even then, however, families moved between *fazendas* for a variety of reasons, from poor housing to personal quarrels with the owners. Bernard Siegel, in a study of migration in Ceará, notes that the region was home to an 'endemic semi-nomadic pattern,' and that "migration from one plantation to another is almost incessant, with an average annual turnover ... of 8 percent."⁵⁶

This feature of life in the *sertão*, the constant recourse to migration as a solution for environmental or labor difficulties, can help us contextualize the "manmade dimension" of the droughts in the late 19th and early 20th centuries. State measures such as an investment in infrastructure could have mitigated the effects of the great droughts, but during the nineteenth century, they were only sporadically applied. Mike Davis writes that "the modern history of the *sertão* is striking for the absence of any significant state developmental role until the 1960s."⁵⁷ Projects such as road building or the construction of dams were occasionally implemented during drought years but then abandoned in years of plenty. While they provided necessary income that allowed some individuals to survive the worst years, they amounted to a stop-gap rather than a robust or long-term solution.⁵⁸

At the coming of the unusually severe 1877 drought, then, the government's minimal investment in public works and relief shipments were woefully insufficient. People started to leave the place of hunger, and the regional cities swelled with refugees: in just two years Fortaleza and Aracati, two cities on the coast, swelled from 25,000 and 5,000 residents respectively, to 114,000 and 60,000. This did not alleviate the need for food, however, and so

⁵⁵ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 253

⁵⁶ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 243

⁵⁷ Davis, *Late Victorian Holocausts*. 398

⁵⁸ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil."

"[a] government pressed on all sides by masses they could not feed decided that the best alternative was to ship thousands of the drought fugitives to the Amazon where they could start life over."⁵⁹

Why the Amazon? Rubber was booming during the worst years of drought, but it was not the only industry recruiting workers. Coffee, grown on plantations in the southeast of Brazil, was similarly booming and needed workers, but they refused to recruit the climate refugees from the *sertão*. Davis notes that "southeastern coffee planters, for their part, wanted only 'white' overseas laborers after Emancipation," and even in the drought stricken regions of the northeast, "[w]hile state authorities were roadblocking *retirantes*' route to the cities and forcibly interning them by the thousands in camps, they continued efforts to lure European immigrants with expensive subsidies."⁶⁰ He describes this policy as part of an obsession among Brazil's elites with "Whitening" and "de-Africanizing" Brazil, which led to the creation of "the mother of all dual labor markets." The racial division of Brazilian society was a fundamental structure of the rubber boom, in which racialized groups were pushed away from the flourishing towns of the coasts and the rich plantations of the south and into the Amazon where free labor was at best a formality.

Rubber Tappers

The rubber tappers of the Amazon have occasioned much scrutiny. In the first part of the twentieth century, attention was paid to the inefficiency of the wild tapping system and the abject working conditions of the tappers themselves.⁶¹ Early on, the rubber trade was placed in a global

⁵⁹ Brooks, "HUMAN RESPONSE TO RECURRENT DROUGHT IN NORTHEASTERN BRAZIL." 43

⁶⁰ Davis, *Late Victorian Holocausts*. 403

⁶¹ For the first see Akers, *The Rubber Industry in Brazil and the Orient*; for the second see W. E. Hardenburg, *The Putumayo: The Devil's Paradise*, ed. C. R. Enock (T. Fisher Unwin, 1912).

context, as scholars, diplomats, and industrialists like tried to understand the different trajectories of the South American and Asian rubber industries, as well as how attention from the countries of North America and Europe could shape their futures. In the last 40 years, scholarship on the workers of the Amazonian rubber trade has shifted towards micro-economic and local considerations, and a re-evaluation of the infamously brutal working conditions. Scholars such as Barbara Weinstein have demonstrated that within the Amazonian system of labor, "the *seringueiro* [tapper] was not necessarily as helpless or dehumanized as one might believe from the standard accounts. On the contrary, the tapper showed himself ready to resist innovations which threatened to infringe further on his autonomy."⁶² We should not take this too far, however, for although it is true that rubber tappers had substantial freedoms when compared to those living under slavery or the compulsory labor schemes of the nineteenth century, they were far from autonomous producers. Rather they labored in a system that offered them superficial legal labor rights while trapping them in a cycle of debt bondage. Despite this, due to the isolated nature of rubber tapping they had significant leverage over key aspects of production that they could exercise to their advantage. This limited power enjoyed by the tappers played a crucial role in shaping the Amazonian rubber industry during and after the boom.

For Barbara Weinstein, the key to understanding the nature of the Amazon rubber trade, and especially its unique characteristics when compared to other Latin American export booms, lies in an analysis of the productive relations internal to the rubber industry. Writing in the 1980s, she criticizes the work of dependency theorists in Latin America for "relegat[ing] such issues as internal class formation ot a position of secondary importance," arguing that we must "look first

⁶² Barbara Weinstein, "Capital Penetration and Problems of Labor Control in the Amazon Rubber Trade," *Radical History Review* 27 (1983). 132.

at the relations of production and exchange that defined the regional economy, and the forces inhibiting any attempts to transform those relations."⁶³ She characterizes her approach as "the exact opposite point of view" of dependentistas who prioritize "production for the world market, rather than the mode of production as the crucial variable." In her analysis, the local productive relations served as a bulwark against the intrusion of capital, as "[e]ven the most heavily financed attempts to transform rubber production and place it on a more 'capitalist' footing had to contend with internal resistance from both the rubber tappers who produced the export commodity and the domestic middlemen who marketed it locally."⁶⁴ Her empirical work into the local conditions of production is invaluable, but her analysis is limited by an insistence on 1) the analytical separability of the local mode of production and the contours of global capital flows, and 2) the primacy of the first over the second. In this section I insist on the co-constitution of local and global manifestations of capitalism - and the inextricability of production and exchange. The local relations of property and the conditions of class struggle in the Amazon did not serve as a bulwark against capital's intrusion, but were themselves fundamentally transformed to serve capital's need not merely for the production of commodities, but their consumption as well.

Central to Weinstein's argument is an insistence on the continued presence of "pre-capitalist relations of production."⁶⁵ Despite the transformation of the Amazon by the rise of the automotive industry's need for rubber, in her view "the Amazon's increased integration into the world capitalist system did not generate a genuine transition to the capitalist mode of

⁶³ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 2

⁶⁴ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 3

⁶⁵ Weinstein, "Precapitalist Relations."

production."⁶⁶ Rather, the rubber industry continued to be dominated by the collecting expeditions that had defined the colonial era. Because of the dispersed nature of resources in the Amazon, their exploitation depended on a similarly dispersed labor force that collected the desired goods on long expeditions into the jungle. The rise of rubber both intensified and reinforced these patterns of production.

Rubber tappers belonged to a complicated hierarchy of tappers (*seringueiros*), their bosses (*patrões*), landlords (*seringalistas*), middlemen and traders (*aviadores*), and exporters. The tappers would be responsible for several looping trails that connected one or two hundred rubber trees. They would collect rubber from these trees, process it, and deliver it on a regular basis to either the local landowner or the trader who managed the local market. The trader is the key figure in the structure of Amazonian rubber production, as one (or several) stood as intermediaries between the rubber collector and the outside market. The trader stationed in the interior served two functions: they collected the rubber from the workers, but they also "operated a wet and dry goods store in one of the tiny interior towns and acted as an agent for a large commercial house, which supplied him with the wares he sold on credit, and accepted as payment the rubber he received either from the patrão [boss] or directly from the tapper."⁶⁷ The trader would generally advance to the tapper the tools necessary to tap rubber, as well as foodstuffs and necessary domestic goods - all at an inflated price.

Thus would form the notorious debt relation that bound the tappers to their positions, and forced them into an endless cycle of work to pay off an ever-increasing debt. The provision of goods and tools was essential, especially for laborers from outside the Amazon who were

⁶⁶ Weinstein, "Capital Penetration." 121.

⁶⁷ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 18. This description of the rubber trade's hierarchy is drawn from pages 15-31.

unaccustomed to the pattern of life on rubber expeditions. For those familiar with the terrain, the jungle provided such foodstuffs as Brazil nuts or cacao beans which could be eaten or sold, but even Indigenous communities had been disrupted to the point that they became reliant on goods provided by the traders. Weinstein says of a particular community in the Amazon called the Mundurucú that "[p]articipation in the trade did eventually destroy much of the tribal culture, but this was due largely to the corrosive effects of the merchandise the Mundurucú received in exchange for their rubber. Increasingly dependent on commodities that they themselves could not produce, members of the tribe spent more and more time on the rubber trails."⁶⁸ The commodities themselves were not locally or even nationally produced, but were largely imported. Resor notes that "the seringueiro subsisted largely on imported canned food, and most of his tools were also imports."⁶⁹ Through the debt contract, the traders connected the rubber tappers to the international market in two distinct ways, ensuring that they produced cheap rubber for international industry, and holding them as a captive market for the commodities produced by international canning industries and tool manufacturers.

Weinstein downplays the coercive nature of this debt, however, even saying that the Mundurucú people were 'coaxed' and not 'coerced.' This distinction is problematic as even in the absence of violence, debt or the threat of starvation are more than enough to coerce the hungry to work. As Jairus Banaji asks: "When is a contract 'voluntary'? The answer is, probably never."⁷⁰ Throughout her descriptions of the rubber trade Weinstein is ambivalent about the forces that compel the tappers to work, describing the relationship between tapper and trader as both "not

⁶⁸ Weinstein, "Precapitalist Relations." 64-65

⁶⁹ Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945." 350.

⁷⁰ Jairus Banaji, *Theory as History: Essays on Modes of Production and Exploitation* (Brill Academic Publishers, 2010). 131.

necessarily coercive in any obvious sense," but also liable to "deteriorate into a coercive relationship, with the tapper spending more and more time in the rubber fields at the behest of his merchant *patrão* (boss)."⁷¹

Her ambivalence seems tied to her insistence on characterizing the rubber trade as pre-capitalist, a claim which relies on the fact that the industry "did not generate a transition to a mode of production in which the producers were completely separated from productive resources or compensated by wages."⁷² In other words, in parts of the Amazon the debt relation provided some limited freedoms to the tappers, especially some control over the means of production. She argues that "in a region as vast as the Amazon, escape to another zone of rubber extractions particularly during the early years of the rubber era, seems to have been relatively easy, thereby limiting the effectiveness of debt as a means of control."⁷³ That the rubber fields were sparsely populated and remote also meant that the laborers retained control over their own working conditions. This modicum of control over the conditions of production mean that the tappers were able to resist proletarianization and the consequent loss of control over their working conditions. At the same time, however, the isolation of the worker undermined the possibility of collective action on the part of the tapper and, as the rubber gatherers penetrated more deeply into the jungle and farther from cities, it increased the tapper's dependence on the trader for essential food, tools, and other commodities.

Coomes and Barham challenge Weinstein's view that the social structure of the Amazon rubber industry was anachronistic. Wage labor did not fail to develop because the laborers, through their resistance to proletarianization, halted the transition to capitalism, and the unique

⁷¹ Weinstein, "Precapitalist Relations." 57-58

⁷² Weinstein, "Precapitalist Relations." 56

⁷³ Weinstein, "Precapitalist Relations." 61

structure of Amazon rubber production was not stalled at a inefficient stage of development. Rather, the debt relation that structured labor in the Amazon was a stable dynamic with its own coherent logic that was suitable to the production of rubber as a commodity in the Amazon. Coomes and Barham's analysis differs from previous analyses in two ways. First, they assert that local actors in the industry retained a significant amount of surplus, and second, that the "much-criticized debt-merchandise contract provided the basis for a stable and relatively efficient relationship among the trader, the patron, and the tapper."⁷⁴

These two points have significant consequences. The first would indicate that the failure of the rubber industry in the Amazon to survive the rise of Asian plantations was not due to a lack of local surplus that could be re-invested in production. Some of this surplus was spent in the conspicuous consumption that characterized the boom cities, and some was invested in the extensive expansion of production by opening new rubber trails deeper in the Amazon (rather than intensive expansion with investment into labor-saving technologies). But Coomes and Barham also make the case that tappers themselves retained significant surplus: according to public land records in Iquitos, rubber workers and their families owned a fair amount of property in the city. Because of the unsupervised nature of rubber collection, tappers could also retain surplus through illicit trades. After providing enough rubber to their contracted trader to service their debt, they could sell any remainder on the side to roaming traders who were looking to purchase any surplus rubber. All told, the returns on rubber tapping were great enough to push that the daily wage in the urban areas of the Amazon succeeded that in other parts of Brazil by a factor of four or five. These city wages were "necessary not only to attract trained labour into the

⁷⁴ Coomes and Barham, "Amazon Rubber Boom Revisited." 241.

region (e.g. professionals, administrators, etc.), but also to hold untrained labour out of rubber extraction to work the ports and service industries of the cities."⁷⁵

The second assertion counters the characterization of Amazonian production as backwards or inefficient. Rather, Coomes and Barham convincingly argue that the structure of the debt relations "ensured the lowest-cost, most sustainable delivery of wild rubber to lenders and estate owners in the decentralized Amazonian environment."⁷⁶ In their view, the debt relation was not a one way siphoning of surplus, but a "mutual commitment of credit for rubber" by which "patron and tapper met their respective needs and ensured that wild rubber would be procured at low cost per unit capital and per tree."⁷⁷ They go further than Weinstein in stating that not only did the tapper retain some leverage in the relationship with the trader, but that the "[t]he debt-merchandise contract filled the common need of trader, patron, and tapper for a stable, long-term relationship in which the trader and patron minimized transaction costs and risks, and the tapper obtained some insurance and access to credit in a risky work environment."⁷⁸ Wage labor, which requires the close monitoring of workers, would have been too costly in the wild rubber scenario where labor productivity (the amount of rubber collected per unit-time) was low, and piece-rate labor would run into "the problem of provisioning the tappers for a season of work."⁷⁹ The debt relation resolved both of these problems. Monitoring laborers was unnecessary because the debt had to be repaid and rubber procured in order to access the commodities necessary for survival in the Amazon. Likewise, the cost of provisioning was offloaded onto the tapper as debt.

⁷⁵ Bradford Barham and Oliver Coomes, "Wild Rubber: Industrial Organisation and the Microeconomics of Extraction During the Amazon Rubber Boom (1860-1920)," *Journal of Latin American Studies* 26, no. 1 (1994). 68.

⁷⁶ Coomes and Barham, "Amazon Rubber Boom Revisited." 245.

⁷⁷ Barham and Coomes, "Wild Rubber." 63.

⁷⁸ Coomes and Barham, "Amazon Rubber Boom Revisited." 245

⁷⁹ Coomes and Barham, "Amazon Rubber Boom Revisited." 246

This debt served as a sort of insurance both for the tapper and their trader. On the one hand, the exorbitant prices charged by the trader dispersed some of the risk bound up in advancing provisions, while at the same time ensuring that the trader was invested in the continued well-being of the tapper. If the loan was to be recovered, the tapper had to be provided with the means of survival.

The analyses of Weinstein and of Coomes and Barham are incomplete, especially on the question of coercion in the relationship between tapper and trader. Coomes and Barham's picture of the debt-merchandise relation as a symbiotic one falls flat in view of the high mortality the tappers faced. Meanwhile Weinstein's distinction between coercion and 'coaxing' is vague and unconvincing. Labor conditions in the Amazon, especially for the migrants from the northeast, were brutal as laborers faced disease, starvation, sweltering weather, and an unwelcoming terrain. Each section of the rubber workers - the Indigenous Amazonians, the peasants from Ceará, and the formerly enslaved Africans - had been cut off from their previous methods of survival. They may not have been forced to work under direct threat of violence, but the possibility of death hung over their heads all the same. Their situation is precisely that described by Marx in *Capital* Volume 1:

[T]hey are compelled to sell themselves voluntarily.... The silent compulsion of economic relations sets the seal on the domination of the capitalist over the worker. Direct extra-economic force is still of course used, but only in exceptional cases. In the ordinary run of things, the worker can be left to the 'natural laws of production', i.e. it is possible to rely on his dependence on capital, which springs from the conditions of production themselves, and is guaranteed in perpetuity by them.⁸⁰

The tappers had precisely the freedom of the proletariat: the freedom to work or starve.

⁸⁰ Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976).

The fact remains, however, that the rubber tappers were not, strictly speaking, wage workers. If wage labor was to be established in the rubber industry, it would need to concentrate both labor and the rubber trees in a smaller geographic area where labor could be supervised and its productivity could be increased. Several attempts were made to establish plantations in the Amazon, especially after the boom as production in the region lagged behind the Asian plantations, but they were largely unsuccessful. The next section will look at the most infamous of these attempts: Henry Ford's Brazilian plantations of Fordlandia and Belterra.

Fordism and the Failure of Plantation Rubber

Almost all rubber produced in the Amazon, both during the boom and in the years after, was harvested from wild rubber trees. Plantations were rare, and to the extent they existed they were small projects "generally regarded as casual experiments by their owners, and were promptly abandoned when problems arose."⁸¹ The main exception to this was Henry Ford's attempt to produce plantation on a large scale in the 1920s and 1930s. His two plantations, at Fordlandia and Belterra, though unsuccessful and eventually abandoned were anything but casual. Over the course of two decades, Henry Ford spent millions of dollars in a Sisyphean struggle to produce rubber in Brazil using management principles perfected in his Michigan factories.⁸² But every time success seemed to be within reach, catastrophe struck and the boulder tumbled down the hill again. His project was stymied by labor unrest and ecological disaster, and in the end his attempt to transplant the relations of production and reproduction that he pioneered in America to the Amazon was a total failure.

⁸¹ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 33

⁸² Greg Grandin, *Fordlandia: The Rise and Fall of Henry Ford's Forgotten Jungle City* (Metropolitan Books, 2009).

However it was, for us, an instructive failure. Henry Ford, fresh off his success in reconfiguring not just the factory floor, but the private lives of his workers, correctly perceived that there was a strong connection between the manner in which workers lived and the manner in which they produced. His mistake was in thinking that this connection was absolute rather than modulated by such factors as ecology, culture, and politics. As we have noted, Henry Ford's reconfiguration of patterns of production was tied up with a project of Americanism - an attempt to elaborate a new kind of man that Gramsci notes was uniquely possible in the United States.⁸³ Existing cultures and communities in the American Midwest had been wiped out through centuries of displacement, resettlement, and genocide and the slate of the US American working classes had been blanked and prepared as a receptacle for Henry Ford's American ideals. In Brazil, just as in Europe, cultural patterns and social structures persisted - they had been changed by the arrival of capital, but not erased. Thus, while correct in his intuition that capitalists in the Amazon would need to pay attention to questions of education, food, child-rearing, demography and recreation, Ford was wrong to import US American answers wholesale.

The factors contributing to Fordlandia's collapse can be grouped roughly into two categories: ecological difficulties exacerbated by Ford's implementation of factory-based regimentation in the Amazonian context, and labor troubles arising likewise from Ford's strict regimentation of the daily lives of his workers according to his experiences in Michigan. This first difficulty - that of the hostility of the Amazonian ecology to the establishment of rubber plantations - did not exclusively affect Fordlandia, but undermined the fundamentals of plantation economics. In the Amazon, rubber trees grow alongside their predators: a fungus that

⁸³ Antonio Gramsci, "Americanism and Fordism," in Antonio Gramsci, *Selections from the Prison Notebooks*, trans. Hoare Quintin and Geoffrey Nowell Smith (International Publishers Co, 1971).

causes South American leaf blight, and a whole host of insects that feed on various parts of the rubber trees. In the wild, rubber trees grow far enough apart that the spread of these predators is limited, and affected trees are less likely to spread their parasites to other hosts. In plantations, however, the close planting of trees create a prime breeding ground for the enemies of the rubber plants.

It was assumed that such horticultural problems would be easily defeated. In the trade journals of the 1920s, proud rubber men declared that "Rubber culture has now been reduced to a science. Tropical diseases have been conquered through proper sanitation. When Mr. Ford enters on his task, therefore, of turning Brazilian jungles into rubber plantations, he will be equipped by modern science to overcome automatically many of the unfavorable elements which brought disappointment and failure to his predecessors."⁸⁴ But Ford's factory science was not up to the task of rationalizing Amazonian ecology. Greg Grandin details the methods used to combat insect infestations and fungal infections implemented by the staff at Fordlandia and Belterra, which were so labor intensive as to seriously undermine the profitability of plantation rubber in South America. These included manual culling of insects from the plants by the men and women of the plantations and regular inspections of the plants for fungus. Eventually Ford's scientists created a system of grafting that combined "durable roots, a high-yield trunk, and a full, verdant crown of blight- and bug-resistant leaves" from three different strains of rubber trees, but "[t]he procedure was time-consuming and costly and entailed building bulky scaffolding in the field to perform the operation and to support the graft."⁸⁵ Even if the management of labor had presented

⁸⁴ "Labor Is Seen as Principal Obstacle to Ford in Brazil," *The Rubber Age* (New York) 22, no. 3 (1927). 125

⁸⁵ Grandin, *Fordlandia*. 330

no difficulties, these measures put Amazonian plantations at a significant disadvantage against Asian plantations where the rubber pests were absent.

The management of labor in the Amazon in fact presented quite a few difficulties for Henry Ford. In an essay in the collection *Imperial Debris*, Grandin describes the current state of Fordlandia where the "hospital has collapsed, the jungle has reclaimed the golf course and tennis courts, and bats have taken up residence in houses where American managers once lived...."⁸⁶ He notes with great irony the similarity between the ruins of Ford's city in the jungle and the urban decay of Detroit, or of Ford's abandoned mining and logging towns in Michigan's upper peninsula. But whereas in Michigan the amenities provided by the company - the high pay, the parks, the healthcare - were a crucial element in the decades-long success of Ford's social experiment, in the Amazon their restrictive nature was more readily apparent. The rubber workers rejected the claustrophobic, managed life that Ford offered them at nearly every turn.

At first, Ford's social policies were welcomed. A hagiographic profile of Henry Ford published in a Brazilian paper in 1928 called Ford a "creative spirit" who was "renewing the face of the planet."⁸⁷ He was expected to bring benefit not only to the captains of industry, but to the whole of society, and workers would flood his plantation to receive the benefits of working for Ford: higher salaries, quality housing, good schools, a state-of-the-art hospital, and ample recreational facilities. Workers did flood the plantation, but found Fordlandia to be far from a worker's paradise.

⁸⁶ Greg Grandin, "Empire's Ruins: Detroit to the Amazon," in Ann Laura Stoler, ed., *Imperial Debris: On Ruins and Ruination* (Duke University Press, 2013). 110

⁸⁷ José Carvalho, "Henry Ford," *O Paiz*, June 14, 1928. "Mas o que é certo é que Henry Ford é um espirito verdadeiramente creador e que está renovando a face do planeta."

Other than some settlements and holdings across the river, Fordlandia was an isolated city. To attract stable workers with families, the town had to provide everything its residents might need. Fordlandia was equipped with radio, a film projector, parks and other perks, but they were all provided on Ford's terms in a way as to promote Ford's family values. In practice, almost every benefit offered by Ford was accompanied by a corresponding reduction in freedom. Housing was one of the most important improvements lauded by Fordlandia's boosters in the press, but it was slow to come, and initially only the Americans in the town lived in permanent housing. These Michigan-style homes were high quality, but were built from materials shipped in from the US, and were designed for a colder climate. Henry Ford evidently detested the look of the thatched roofs and mud walls on the homes workers had built for themselves, considering them unbecoming of his workers. By the early 1930s as the construction of housing expanded, the residents of Fordlandia were encouraged to abandon their well ventilated housing for the stifling heat of metal-roofed houses, described as "galvanized iron bake ovens."⁸⁸ In a program similar to the Five Dollar Day, which increased the pay of workers who met Ford's lifestyle requirements, the extra rental cost of the new houses was "not actually paid by the laborer inasmuch as his salary [was] increased by that amount in the event that he live[d] in one of these houses."⁸⁹ Despite this, many of the married workers eligible for the 'modern' houses "prefer to remain in their huts by the river - palm thatched shelters with dirt floors - in preference to the frame houses with screened windows, running water, bathrooms and electric lights."

The food and recreation offered by Ford were likewise rejected by Brazilian workers. Drinking was banned in Fordlandia but the settlements across the river were outside his

⁸⁸ Grandin, *Fordlandia*. 274

⁸⁹ A. Ogden Pierrot, "A Visit to Fordlandia," *The Rubber Age* (New York) 31, no. 1 (1932): 19–22.

jurisdiction and they grew into "'filthy small cafes, restaurants, meat and fruit shops,' gambling houses, and thatched bordellos established by local merchants and staffed mostly with women from Brazil's poor northeast."⁹⁰ Rather than spend time listening to music or visiting the parks of Fordlandia, the workers preferred to wind down off premises across the river. When they ate on site, the workers were provided food in an eating hall, but they "grumbled about being fed a diet set by Henry Ford, consisting of oatmeal and canned peaches imported from Michigan for breakfast and unpolished rice and whole wheat bread for dinner."⁹¹ Adding to their frustration, the cost of meals were deducted automatically from the worker's paychecks. In December 1930, the workers rioted in the cafeteria, and the frustrations of the compound's restrictions boiled over as they ransacked the town. In response the company fired all but a few hundred men, and they were forced to renew their search for a suitable mass of workers free from "shiftless and vicious types" and "communistic elements."⁹²

Even the hospital - maybe the most successful of Fordlandia's social improvements - was a source of conflict. Boosters in trade publications such as *India Rubber World* and *The Rubber Age* lauded the eradication of diseases endemic to the Amazon, even going so far as to claim that Ford's settlement "is probably as free from malaria and other tropical scourges as any American city."⁹³ Indeed the hospital built at Fordlandia provided care not only to employees and their families, but to anybody from the neighboring districts. The introduction of health measures were largely successful, but residents bucked against the injunction to give birth at the hospital, rather than at home with a midwife. Ford's doctors also insisted on providing mothers and new children

⁹⁰ Grandin, *Fordlandia*. 157

⁹¹ Grandin, *Fordlandia*. 228

⁹² Pierrot, "A Visit to Fordlandia." 20

⁹³ "Ford Rubber Plantation," *India Rubber World* (New York) 85, no. 5 (1932). 48

with powdered milk - which meant that "utensils had to be provided as well, since most workers didn't own dishes, or 'even a spoon,' to prepare the powdered milk, using instead their fingers to mix the powder in empty cans."⁹⁴

Common to all of the restrictions Ford placed on his workers was the substitution not merely of American culture for Amazonian practices, but the substitution of American commodities for local materials. For Henry Ford the ideal worker is one who not only produces commodities, but consumes them as well. In the early years of the plantation, the establishment of a cash system with nowhere to spend it undermined Ford's dream of a modern (and teetotaling) Midwestern city in the jungle. To rectify this, cash wages had to be combined with a robust domestic cash market. Ford attempted to manage the whole of the circuit of commodity production and consumption on site - by 1932, Ford's Amazonian cities were homes to rows of shops. Central to Fordlandia's propaganda was the image of domestic commodity consumption:

The housewives who for the first time in their lives know the joys of shopping are to be seen moving from store to store and finally emerging with market baskets of conventional design filled with groceries. This is a considerable transformation from the erstwhile occupation of these same women of gathering a few bananas from the small plot behind the hut on the edge of the jungle and serving them with whatever fish the male members of the family may have caught....⁹⁵

But these small glimpses of midwestern life being lived in the Amazon were not generalized to a large Amazonian rubber working class. They were a mirage, carefully reproduced as propaganda for the trade journals of the rubber industry. This was a facsimile of Midwestern industrial life, and it was not attractive to workers in Brazil. They chafed at the closely managed routines of Fordlandia, and although they came to Fordlandia in large quantities, they mostly did not stay

⁹⁴ Grandin, *Fordlandia*. 291

⁹⁵ Pierrot, "A Visit to Fordlandia." 22

long enough to see the rubber crop through: "A fair number disappeared after their first pay day with more money than they had ever seen at one time, and after more than two years have failed to put in another appearance."⁹⁶

By the early 1940s, some 15 years after Ford purchased land in Brazil neither Fordlandia nor its sister plantation Belterra were producing rubber in commercial quantities. Pest and disease management was expensive, which along with the higher cost of labor in the Amazon made competing with cheap Asian rubber difficult. The biggest issue, however, was labor recruitment. Despite the amenities and higher wages offered at the plantation, Ford was unable to attract sufficient labor to Brazil. In an article examining the difficulties facing Fordlandia and Belterra in 1942, Joseph A. Russell notes several factors exacerbating the labor supply problems. Prominent among them were the "disinclination to work on the part of the natives who are without employment," and the "bad reputation among Brazilian labor" of the rubber region of the Amazon.⁹⁷

The Second Rubber Boom: Rubber Soldiers

Periodically, after the boom years, discussion would arise of a rebirth of Brazilian rubber. In 1916 Joseph F. Woodroffe published *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained*, a lengthy book which set the conventions of the genre: despite the rise of plantation rubber in Asia, with the right tweaks to both production and the reproductive conditions of the workers, Brazilian rubber could still be profitable.⁹⁸ His

⁹⁶ Pierrot, "A Visit to Fordlandia." 21.

⁹⁷ Joseph A. Russell, "Fordlandia and Belterra, Rubber Plantations on the Tapajos River, Brazil," *Economic Geography* 18, no. 2 (1942). 140.

⁹⁸ Joseph F. Woodroffe, *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained* (T. Fisher Unwin, 1916).

suggestions primarily centered on developing a stable working population, to be achieved through import of laborers from Asia, as well as replacing the dominance of rubber collection with an agricultural system that combined rubber with crop and livestock cultivation. In his vision, the imported labor would "clear and settle on the lands available up the Amazon, and whilst forming home-steads and small farms, to tap and cure rubber to obtain 'pocket-money,' whilst their women and children do the lighter work on the home farms."⁹⁹ A population of small proprietors would be significant change from either plantation development or the existing system of debt-bound tappers, but Woodroffe was insistent that the self-sufficiency of tappers would improve the long-term stability of the rubber industry in Brazil.

Cognizant that the revival of Brazilian rubber could be seen as a challenge to British dominance in the rubber trade, Woodroffe invited both former president Roosevelt and Lord Bryce in England to comment for his book on the relevance of the Monroe doctrine to British investment in Brazil. Both declined to say anything of substance on the matter, and indeed in later discussions Brazilian rubber was presented as a crucial matter of national interest for the United States. After all, in the 1920s, "[w]hile British companies in British colonies produced 67 per cent of the world's rubber, most of their exports went to the United States, which consumed over 70 per cent of the world's production."¹⁰⁰ This situation came to a head in early 1942 when Japan captured Singapore from the British and cut off the Allied powers from their supply of natural rubber. Almost overnight, the United States was deprived "of 97% of its supply of a key material for which it had effectively no domestic sourcing."¹⁰¹ Shortly thereafter US chauvinists

⁹⁹ Woodroffe, *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained*. ix.

¹⁰⁰ Randolph R. Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945," *The Business History Review* 51, no. 3 (1977). 361.

¹⁰¹ Alexander J. Field, "The World War II US Rubber Famine," *Business History Review* 99, no. 3 (2025).365.

began to extol the virtues of the long-neglected Amazon as "[t]he loss of Asiatic sources of supply has, of course, turned the attention of the United States to the potentialities of Brazil's rich valley" - this time, of course, the lesson was learned, the mistakes of the past would not be repeated, and "the new phase of the trade [would] follow along economic and constructive lines."¹⁰²

In reality, no such thing happened - rather than investing meaningfully in a stable rubber industry, Brazil fell back on old patterns of extraction. In the 1940s Brazil made one last push to harvest rubber from the Amazon, but the lesson it had learned was less about the investment in stable industry than the disposability of labor. Surveys of the Amazon indicated that the region could supply 100,000 tons of wild rubber per year, but this output "implied, according to the survey's optimistic calculation, 100,000 tappers, to be drawn mostly probably from the arid and impoverished northeast, as in the days of the boom."¹⁰³ The previous system of tapping remained mostly intact, and attempts to bypass it by offering tappers higher prices were countered by the *seringalistas* raising the prices of the provisions that tappers relied on - ensuring that they were able to maintain profits. Further, along the rubber chain traders were hoarding rubber, waiting for prices to rise further, and production was much lower than expected. The Brazilian government decided to abandon their attempts at control, "turn the entire system 'back to the Brazilians'" and focus their efforts on recruiting more labor.¹⁰⁴

Labor, long the primary obstacle to the viability of Amazonian rubber, was not easy to obtain. "The Brazilian government in February 1943 turned the recruitment of tappers from

¹⁰² John Melby, "Rubber River: An Account of the Rise and Collapse of the Amazon Boom," *The Hispanic American Historical Review* 22, no. 3 (1942). 468.

¹⁰³ Warren Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History* (Cambridge University Press, 1987). 93.

¹⁰⁴ Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History*. 95

outside the region into a quasimilitary operation, contracting them for two years, paying advances and promising family allowances, and furnishing transportation."¹⁰⁵ As previously, the benefits were exaggerated, medical services were inadequate, and advances went unpaid. The rubber tappers brought into the region had to battle disease, malnutrition, and mistreatment, with minimal protection from the state. The labor conditions of these 'rubber soldiers' were so bad that as early as 1946, Time Magazine reported that "[o]f the 18,000 men who went into Amazonia, only a few were ever seen again."¹⁰⁶ Sources disagree on the exact number of people conscripted to gather rubber during the war with more recent estimates exceeding 55,000, as many as half of whom died in the jungle.¹⁰⁷ Amazon exports to the US quadrupled during the war period, reaching just over 20,000 long tons in 1944 - just over a fifth of total US rubber imports, and less than 2% of the total preblockade imports of over a million long tonnes.¹⁰⁸ Compared with the US Government's expectations of importing 40,000 long tons in 1944, the Amazon's second rubber boom was a resounding failure.¹⁰⁹

Conclusion

At no point in the history of Amazon rubber - from the heyday of the rubber boom to the failed plantation experiments to the wartime conscription of rubber soldiers - was a body of free wage laborers established. Despite this, the key categories of life under capitalism that seemed so

¹⁰⁵ Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History*. 96

¹⁰⁶ "BRAZIL: Lost Army," *Time Magazine*, July 22, 1946.

¹⁰⁷ Gilson Laone Pereira, "'SOLDADOS DA BORRACHA' - ESQUECIDOS OU NÃO LEMBRADOS?," *Revista Margens Interdisciplinar* 8, no. 11 (2014).

¹⁰⁸ *Report on the Rubber Program, 1940 - 1945* (Rubber Reserve Company, 1945).

¹⁰⁹ Edward D. McLaughlin, "Report by the Consul at Para (McLaughlin) on Changes Brought About in the Amazon Rubber Program as a Result of Recent Conferences in Washington," Foreign Relations of the United States, Diplomatic Papers, US Department of State, 1943.

important for Henry Ford to grab hold of played a crucial role in structuring the lives of the rubber tappers. Every aspect of the lives of workers were bent to the needs of capital accumulation, including the manner in which they reproduced themselves while not working, and the commodities they consumed.

Capital is sometimes seen as a homogenizing force, which by collapsing all labor into abstract labor-time evacuates it of its specificity. The role of wild rubber in the Amazon points to a different interpretation. The exchange relation posits the commensurability of various types of concrete labor, but it does not collapse them. The need of capital for various specific concrete labors flourishes, conditioned by differences the processes of production, differences in political and cultural structures, and differences in the ecological conditions of extraction. Perhaps more to the point, the individuals who provide this concrete labor to capital serve another role as well: they are a crucial market for the commodities they produce. As consumers they are differentiated as well: the wage worker in the automotive factory purchased quite different goods than the rubber tapper in the Amazon, but they both contributed to the essential final stage of commodity production: consumption.

The rise and fall of rubber production provides key insight into the categories of Fordism. Henry Ford recognized that the private lives of his workers were in a real sense internal to the production process. For automotive production to succeed as more than a niche luxury good, cars would have to be produced more cheaply, and the working classes would have serve as portion of the automotive market. This, combined with homeownership and an increase in domestic commodities reduced the flexibility of factory workers and tied them to their towns and jobs. Likewise, in the Amazon, in order for workers to be able to produce rubber cheaply, they would

have to be severed from their previous means of subsistence and posited as commodity consumers. The substitution of consumption for subsistence production served the double purpose of tethering the workers to their employment and providing a market to the producers of canned food and tools abroad.

The debt relation was the dominant labor dynamic in the Amazonian rubber trade, but it was not the only one. In the upper reaches of the Amazon, state presence was minimal and the extraction of quality rubber became more difficult. Under these conditions, quite a different labor dynamic developed to meet the rubber needs of the automotive industry. In the Putumayo region of what was then Peru, but is now Colombia, the agents of the Peruvian Amazon Company subjected the Indigenous inhabitants to a system of slavery and genocide. In the context of different ecological and productive relations, the needs of capital shifted, and human life became comparatively less valuable - less worth reproducing. The next chapter will take a closer look at this particular case.